Fundamental Labs Report Chapter 9

Purpose of the labs: To learn the phenomena of semiconductors and its application to diodes and transistors. In the first week, we learned about diode and its characteristics, and half-wave and full-wave rectification, and constructed diode circuits. In the second week, we learned about transistors and bipolar transistors. We then constructed an amplifier circuit using a bipolar transistor.

Week 1

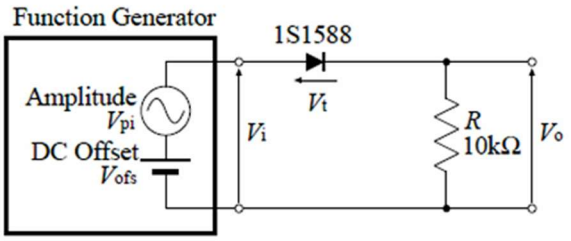
Purpose: To learn about diodes

1.1 Half Wave Rectification

1.1.1 Basic Operation confirmation

Experiment details

We constructed a rectifier circuit. The diagram is below. We used the function generator with the frequency of 1 kHz for input voltage V_i. We then connected V_i to the oscilloscope Ch1and connected the output V_o to Ch2. We selected DC mode, and used Ch1 for SOURCE. For the experiment, assuming a frequency of 1kHz and V_{ofs} =0 V for V_i, we changed V_{pi} from 0.1 Vin the order of xV, 4V, and observe Vo. Here, x first indicates the voltage at which V_o appears.

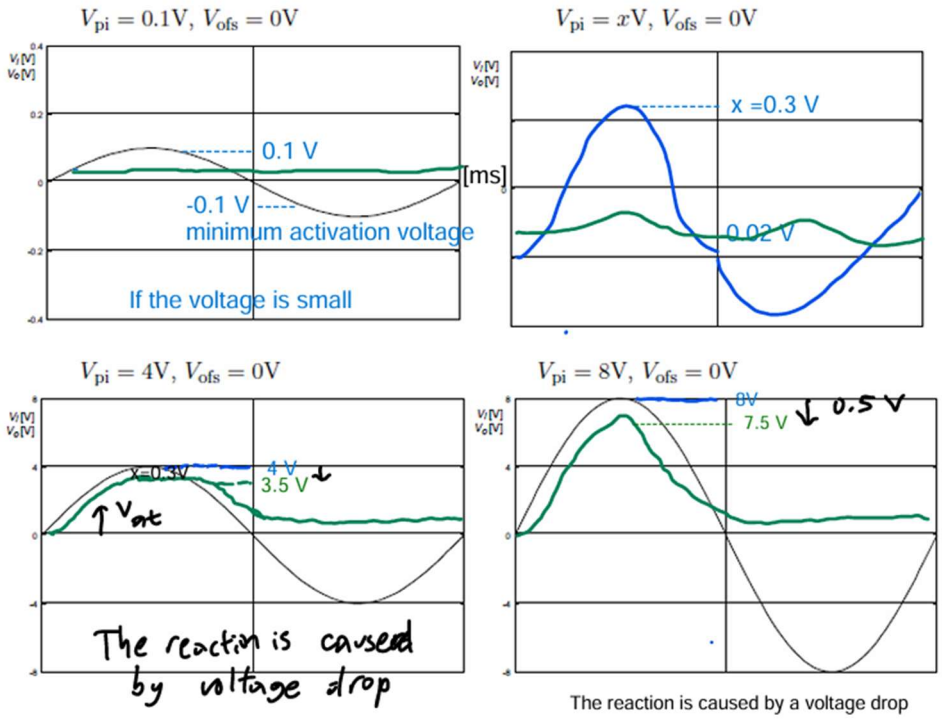


Result

The graph of the result is below. The graph drawn in green is the voltage output (measured in volts), while the graph drawn in black or blue is the voltage input. The output graphs for $V_{pi} = 0.1, x, 4, and\ 8V$ respectively never reaches 0.

For V_{pi} =0.1 V, the minimum activation voltage is -0.1 V.

From the graph, we can see the value of V_{pi} =x V is 0.3 V. Hence, x=0.3 V.
For sufficiently large voltages $V_{pi} = 4\ V$ and 8V, for each of the voltages, the output voltages drop by the value of x. x is the internal resistance. The internal resistor decreases the output voltage by xV.



Question

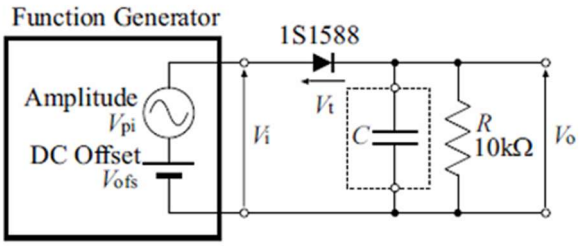
1. Two diode characteristics:
   a) The value of the voltage output after the diode is applied to the circuit never reaches below zero since if the input voltage is negative, the current of the output would be stopped (become) zero so that the output is zero.
   b) From the graph for Assignment 1.1, the difference in amplitude between the input voltage and the output voltage is the internal resistance for all values of Vi from 0.1 V to 8V. x = 0.3 V, and the internal resistance is also 0.3 V
2. From the graph, x V= 0.3 V, and the internal resistance is also 0.3. So the internal resistance is 0.3.

1.1.2 Smoothing capacitor

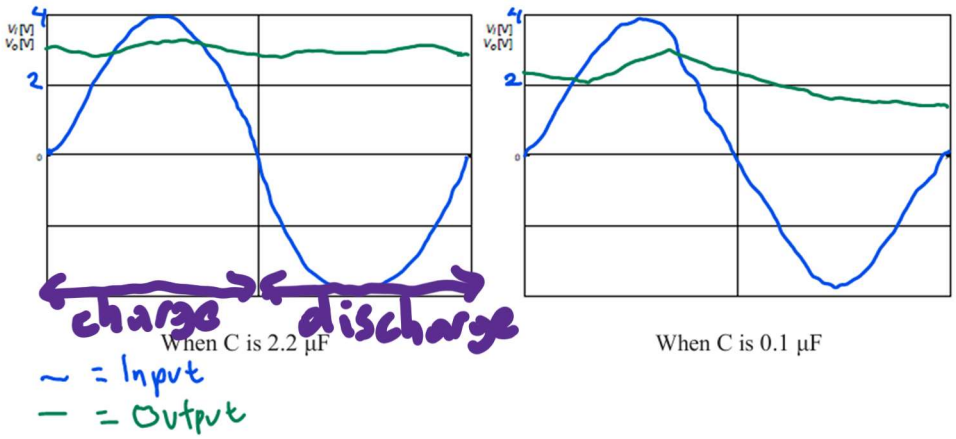
Experiment details

We added a capacitor with values of 0.1 µF and 2.2 µF respectively to the previous rectifier circuit to see the effects of a smoothing capacitor. Assuming V_{pi} is 4V, we varied the capacitor capacitance in both directions to 0.1 µF and 2.2 µF, and observed the change in V_o .

The circuit diagram is below.



Result



The result is the graph above. The voltage drop and fluctuation of V_o is smaller when the capacitor is 2.2 μF compared to when C is 0.1 μF.

Question

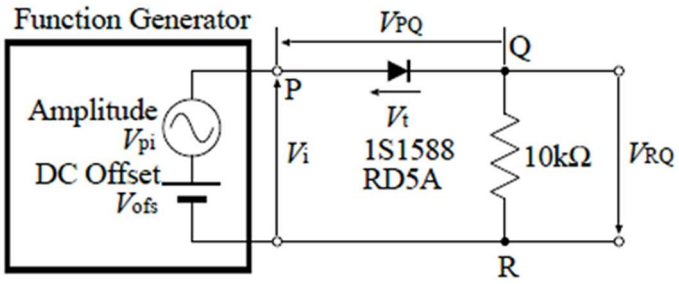
Effect of a smoothing capacitor:

When the input voltage is positive, the capacitor experiences charge and when the input voltage is negative, the capacitor experiences discharge.
From the graph, we can see for the larger capacitor (2.2 microfarad) that there is a smaller voltage drop and smaller fluctuation. The higher the value of the capacitor, the greater the smoothing. Smaller voltage leads to bigger drop due to quicker charge discharge.

1.2 Diode V-I characteristics

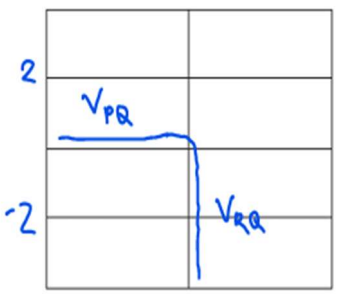
In here, we compare the characteristics of diode 1S1588 with a Zener diode RD5A

We assumed V_i to be a sine wave or triangle wave of V_{pi}=8 V, V_{ofs} = 0V.

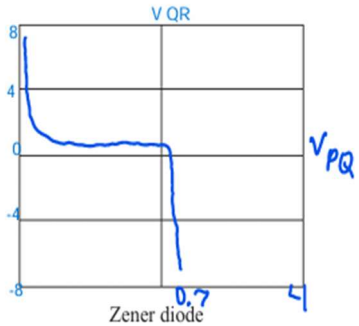


As in the figure, we set three points P,Q, and R and respectively connected oscilloscope GND to point G, the tip of the Ch2 probe to point R, and the tip of the Ch1 probe to point P. We then set the oscilloscope to X-Y display to calculate the amplitude gain from the slope of the xy-line of the graph and enter results. After that. We replaced the diode 1S1588 with a Zener diode RD5A, and draw the X-Y display results. The diode direction is the same as in the figure above.

Result



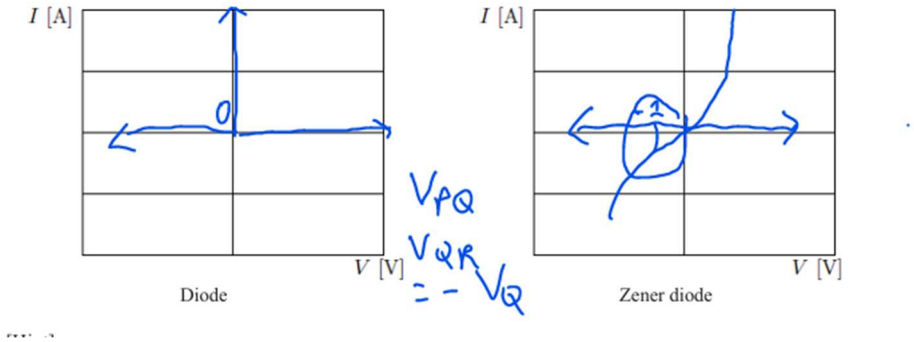
Diode



Zener diode

Question

The V-I characteristics of the diode and Zener diode are drawn in the figure below



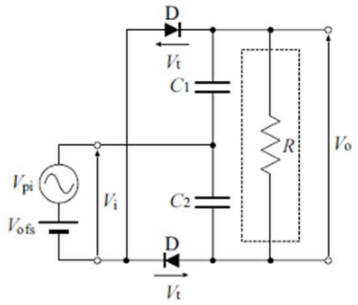
Diode

Zener diode

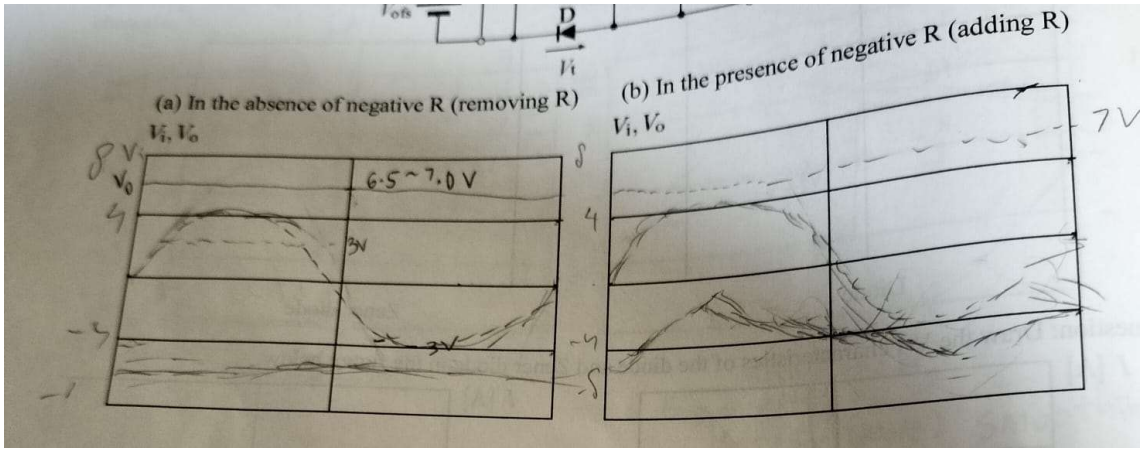
1.3 Full wave-voltage rectifier circuit

This is wired on a breadboard as in the figure below. A 0.1 μF Mylar capacitor is used for capacitors C1 and C2 (labeled 104; This implies 10 × 10 pF). Assuming Vpi= 4V,

we observed Vi and Vo. However, in this circuit, Vi and Vo do not represent a common GND; therefore, it is not possible to simultaneously observe them with a lab oscilloscope (GND for Ch1 and Ch2 is connected inside the oscilloscope). By momentarily observing only Vi, we observed only Vo after adjusting Vpi. In addition, we observed the changes in the waveform when a negative resistance R (= 10 k Ω)) is added.



Result



Question

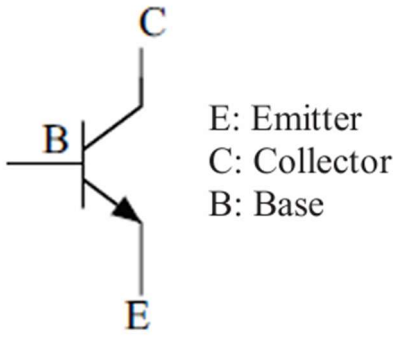
1. Discuss load effects

Load effect is the change in the steady-state value of the DC output voltage or current resulting from a specified change in the load current (of a constant-voltage supply) or the load voltage (of a constant-current supply), with all other influence quantities maintained constant. The fluctuation and voltage drop is smaller when there is absence of R, hence load effect is bigger when there is no R.

Week 2

Purpose: In this week, we learn about transistors and bipolar transistors. In the lab, after first conducting experiments with an npn-type bipolar transistor 2SC1815, experiments are conducted with an n-channel junction-type FET 2SK30A. A circuit is made on a breadboard for the experiments.

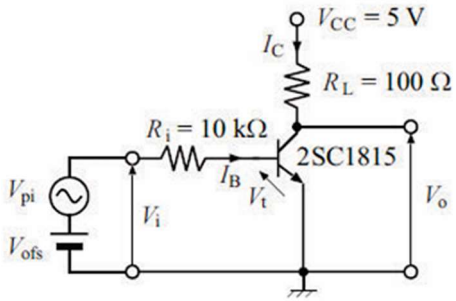
The X-Y display is useful for measuring amplification characteristics. When the amplification factor increases and the oscilloscope is DC coupled, measurement becomes more difficult, so measurements may be performed by using AC coupling in which only alternating current components appear.



2.0 Bipolar transistor basic characteristics

Experiment details

We constructed the following circuit and used the function generator for Vi. We then set

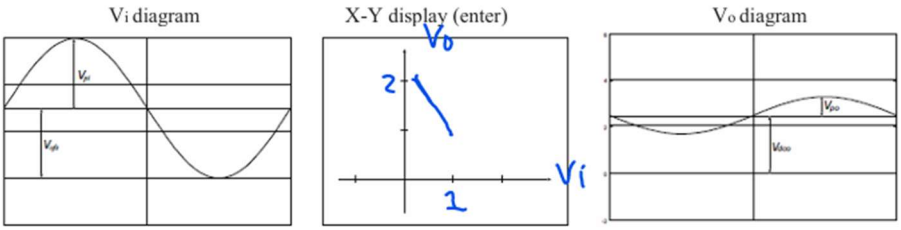


Vcc using a direct power supply. For reading the input voltage Vi and output voltage Vo, we connected Vi to oscilloscope Ch1 and Vo to Ch2.

After setting input Vpi, Vofs so that output is not saturated, we set the oscilloscope to X-Y display and confirm the relationship between Vi and Vo.

Result

We determined the relationship between Vi and Vo and calculated the slope of the graph in the XY-plane to get the amplitude gain.



The slope of the line from the X-Y plane is -2.5 ≈ -2.

Hence, the amplitude gain is A = Vo/Vi = slope =-2

The value of β can be calculated from the equation to determine the amplitude gain:

$$A = \frac{V_o}{V_i} = \frac{I_c R_l}{I_b R_i} = \beta R_L / R_i$$

$$-2 = \beta\left(\frac{100}{1000}\right)$$

$$\beta = -200$$

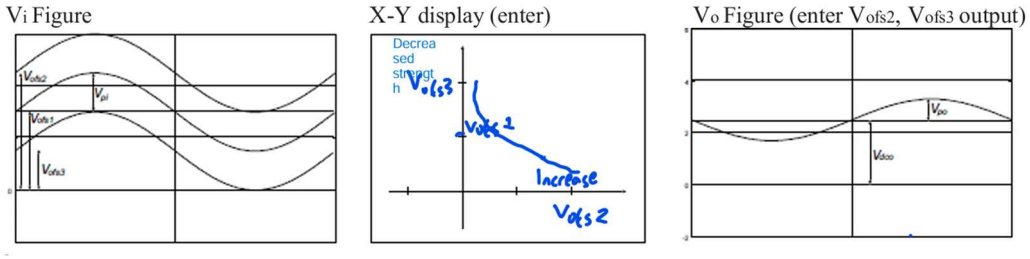
2.1 Amplifier circuit

(a) Operating characteristics when $V_{cc} = 5\text{V}$

Experiment details

We constructed the following amplifier circuit and determined operating characteristics when $V_{cc} = 5\text{V}$

Result



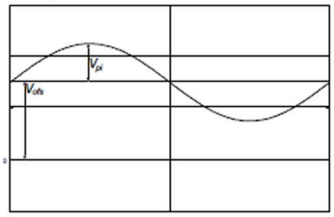
(b) Operating characteristic when varying V_{cc}

Experiment details

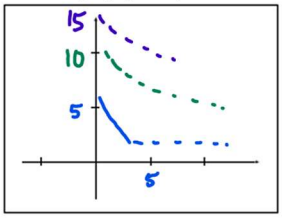
We varied V_{cc} to 5V, 10V, and 15V. We then observed the changes in the amplification characteristics when changing V_{cc} to 5V, 10V, and 15V, respectively.

Result

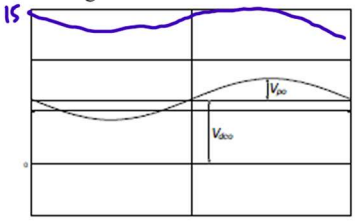
Vi Figure



X-Y display (enter)



Vo Figure



From the graph, we can obtain the relationship that when V_{cc} is increased, the graph is translated upwards and the Vo-Vi lines are parallel to each other with differing height (highest for Vcc = 15V and lowest for Vcc =5V) . Vo is translated upwards as V_{cc} increases.

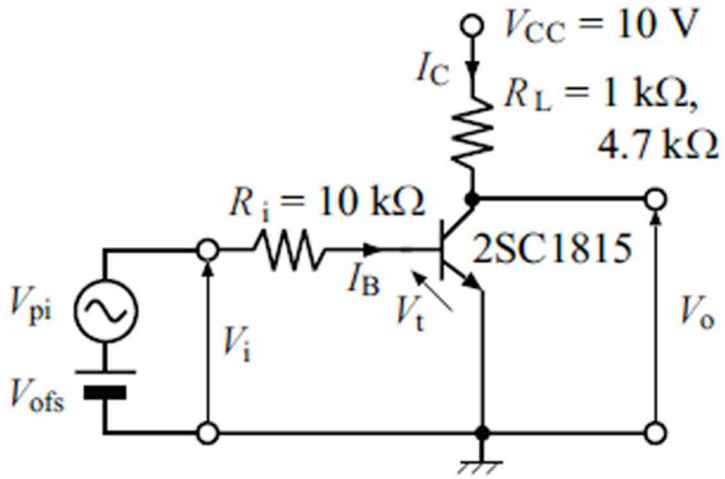
Question

The range of output that is not saturated depends on V_{cc}, and cannot exceed the value of V_{cc}.

2.2 Amplification to obtain large voltage

Experiment details

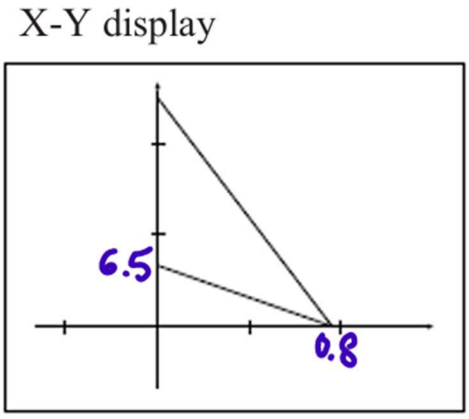
We constructed the following circuit and set Vcc to 10V and changed RL in section 2.1 to 1k ohm.



We then repeated the experiment, only changing RL to 4.7 kilo ohms.

Result

We set the result graph in the oscilloscope and obtained the following slope



The amplification factor when $R_L = 1$ kilo ohms is approximately 23, while the amplification vector when $R_L = 4.7$ kilo ohms is approximately 112.

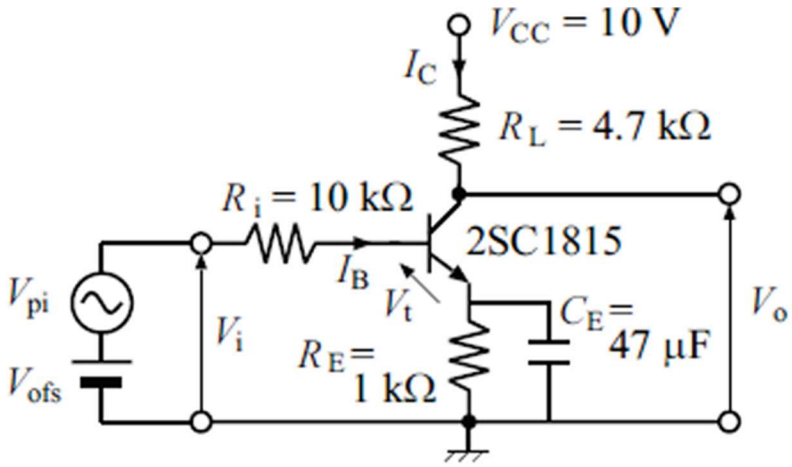
The relationship between R_L and the amplification factor A is the following:

$$A = \frac{V_o}{V_i} = \frac{I_c R_l}{I_b R_i} = \beta R_L / R_i$$

Hence, A is proportional to R_L/R_i .

2.3 Bias current stabilization

Experiment details



As in the figure below, we added RE and CE, and examine the amplification factor (Vpo/Vpi) of the alternating current signal. We then removed CE and compared the result with the case when there is only RE.

Result and Question

Question 1: The purpose of introducing RE is to decrease β and thus decrease fluctuation of I_b. Using RE has a stabilizing effect, using negative feedbac to quickly reach saturation.

Question 2: Adding CE (bypass capacitor) to the bipolar transistor circuit increases the amplification factor Adding CE bypasses RE for alternating current signals, effectively shorting it for AC signals (high-frequency signals). This removes the negative feedback, increasing the gain.

Report

Question 1

The simultaneous equations are as the following:

$$V_c = \begin{Bmatrix} V_i = I_b R_i + V_t \\ V_{cc} = R_L I_c + V_c \\ I_e = I_b + Ic, I_e = \beta I_b \end{Bmatrix}$$

Solvig the simultaneous equations, V_c becomes $V_c = -\frac{\beta R_L}{(1+\beta)R_E+R_i}(V_i - V_t) + V_{cc}$

The equation for the output voltage V_o in terms of V_i is the following :

$V_o = V_{cc} - I_c R_l$

$= R_L I_c + V_c - I_c R_l$

$= V_c = -\frac{\beta R_L}{(1+\beta)R_E+R_i}(V_i - V_t) + V_{cc}$

To calculate the amplification factor:

$$A = \frac{V_o}{V_i} = \frac{I_c R_l}{I_b R_i} = \beta R_L / R_i$$

Question 2

1. The reason for which the input-output characteristics of the grounded emitter transistor circuit are linear:
   Referring to Question 1, the characteristic input and output equation
   $V_c = -\frac{\beta R_L}{(1+\beta)R_E+R_i}(V_i - V_t) + V_{cc}$ is in the linear equation form $y=mx+b$,
   Where m=$\frac{\beta R_L}{(1+\beta)R_E+R_i}$, x=$V_i - V_t, where\ V_t\ is\ constant$ and b= V_{cc}
2. From experiment in Section 2.2, We calculated the amplification factor when RL =1k ohm is approximately 23.

The amplification factor when RL= 4.7 k ohm is approximately 112.

112 divided by 23 is approximately 5

3. Adding CE (bypass capacitor) to the bipolar transistor circuit increases the amplification factor. Adding CE bypasses RE for alternating current signals, effectively shorting it for AC signals (high-frequency signals). This removes the negative feedback, increasing the gain.